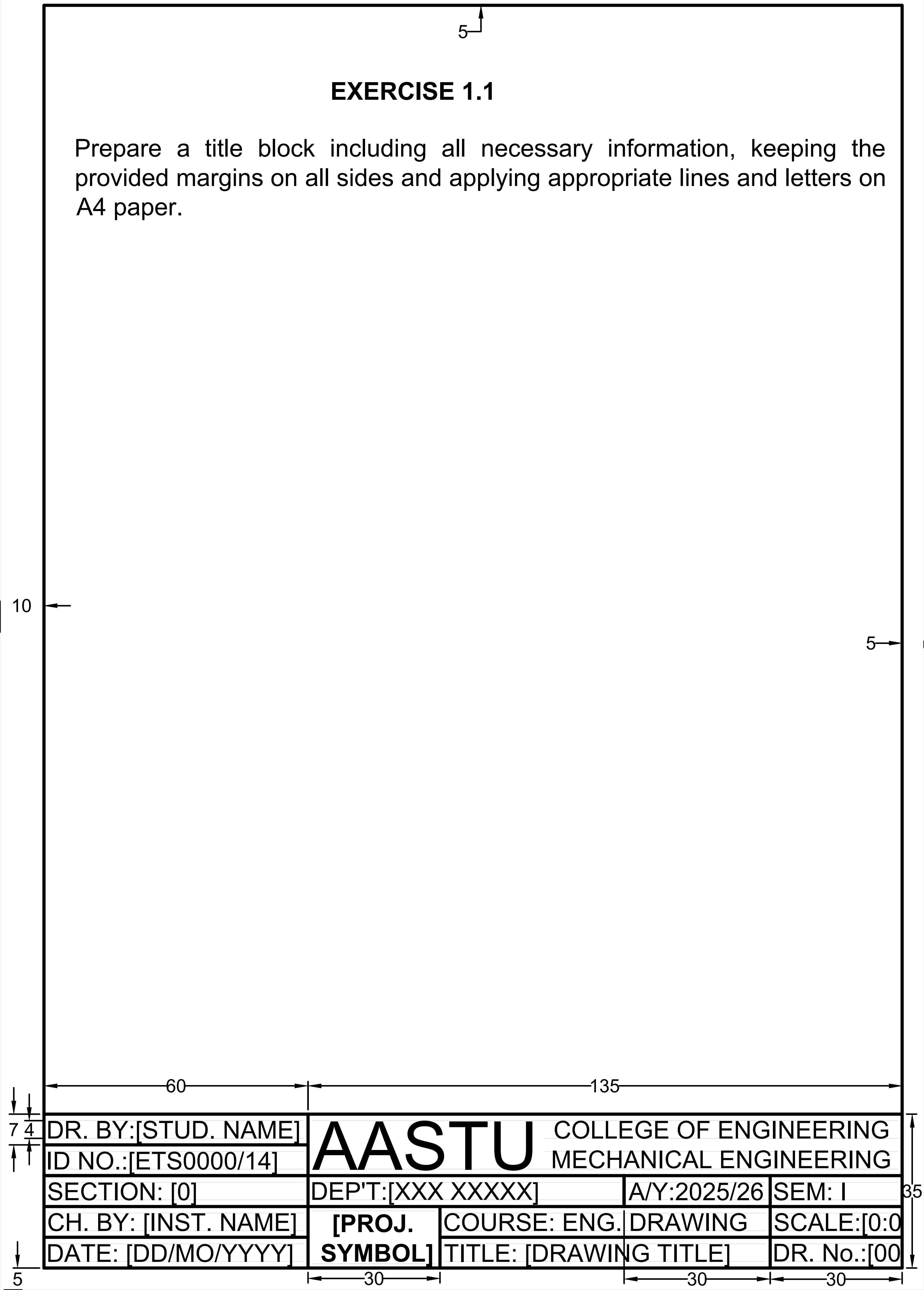
EXERCISE 1.1

Prepare a title block including all necessary information, keeping the provided margins on all sides and applying appropriate lines and letters on A4 paper.

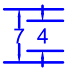


EXERCISE 1.2

Using uppercase letters and numeric heights of 4 mm, single-stroke gothic-style letters, and maintaining appropriate spacing between characters and words, as shown in the examples:

#a. copy the sentences and numbers.

#b. Redraw the guidelines and copy the texts



A GOOD STUDENT REALIZES THE IMPORTANCE OF SPEED AND ACCURACY OF TECHNICAL DRAWING.

1a.

1b.

2. THE QUICK BROWN FOX JUMPS OVER THE LAZY DOG.

2a.

2b.

3. the quick brown fox jumps over the lazy dog.

3a.

3b.

4. 1 2 3 4 5 6 7 8 9 0

4a.

4b.

DR. BY:	AASTU	COLLEGE OF ENGINEERING	
ID NO.:		MECHANICAL ENGINEERING	
SECTION:	DEP'T:	A/Y:	SEM:
CH. BY:		COURSE:	SCALE:
DATE:		TITLE:	DR. No.:

EXERCISE 1.3

Redraw the following lines as show in the examples (#a).

1. Border line (line thickness/weight 0.7mm)

1a.

1b.

1c.

2. Visible/ outline line(line thickness/weight 0.5mm)

2a.

2b.

2c.

3. Hidden line (line thickness/weight 0.35mm for thin and 0.5mm for thick)

3a.

3b.

3c.

4. Center line(line thickness/weight 0.35mm)

4a.

4b.

4c.

5. Dimension line, Extension line, Section line (line thickness/weight 0.35mm)

5a.

5b.

5c.

6. Construction line (line thickness/weight 0.18mm)

6a.

6b.

6c.

DR. BY:
ID NO.:
AASTU
COLLEGE OF ENGINEERING
MECHANICAL ENGINEERING
SECTION:
DEP'T:
A/Y:
SEM:
CH. BY:
COURSE:
SCALE:
DATE:
TITLE:
DR. No.:

AASTU
UNIVERSITY FOR INDUSTRY

COLLEGE OF ENGINEERING

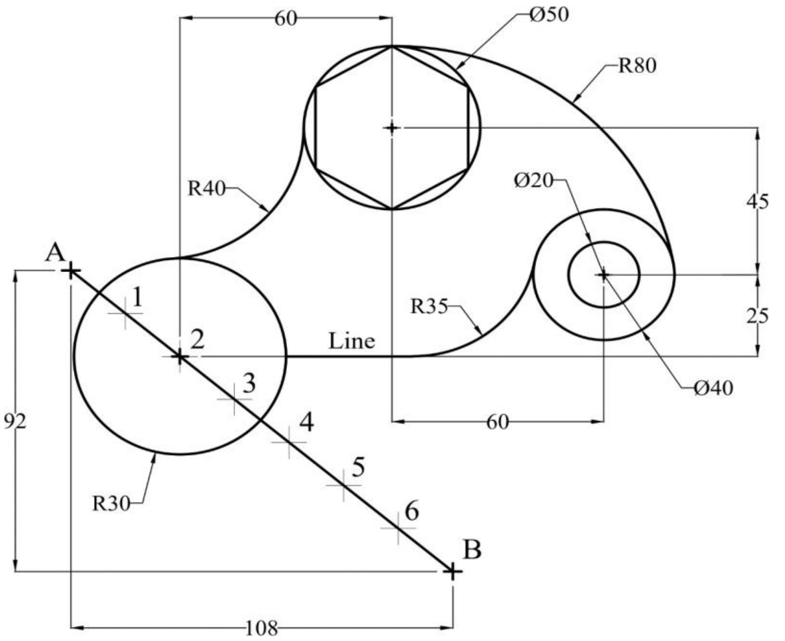
DEPARTMENT OF MECHANICAL ENGINEERING

ENGINEERING DRAWING (MEng2101) A/C Year 2026/27 Semester I

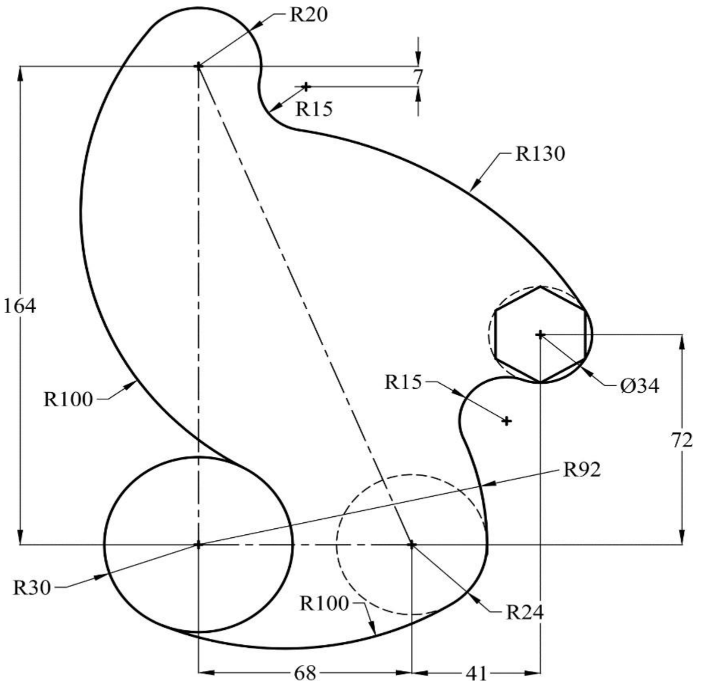
Exercise Set-1

Exercise 1.4. Geometric Constructions

a. Draw an 80 mm long line at an angle of 30° from the horizontal and draw a line parallel to it at a distance of 30mm.
b. Draw a 75mm long line at 15° from the vertical and draw a perpendicular bisector of it.
c. Draw 120° using a straight edge and a compass and bisect it.
d. Draw a triangle whose sides are 70mm, 60 mm, and 80 mm long.
e. Circumscribe and inscribe a regular pentagon in a circle with a diameter of 70 mm.
f. Draw a triangle whose one side is 50 mm and the included angles from this line are 45◦ and 75◦.
g. Draw a regular hexagon if the distance across flat is 60mm.
h. Draw a square with 60mm sides, and inscribe an octagon.
i. Draw two circles whose radii are 30mm and 20 mm with centers 54mm apart. Draw a tangent line to the lower sides of both circles which don't cross the line connecting the center points. Draw a tangent line to both circles that intersects the line connecting the centers.
j. Use the four-center method to draw an ellipse whose major and minor diameters are 76mm and 54mm, respectively.
k. From point (A) and divide the given line (AB) into 7 equal parts, using Diagonal Line Technique. Use point (2) to start drawing the given shape.



l. Draw the given view. Show all constructions and mark all tangents



m. Draw the following, show the construction lines, and mark all tangent points. Find the radius of the circle that passes through points (A), (B), and (O).